

Netflix Engagement Report: A Case Study

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1 Summary

This report analyzes Netflix's first-half 2025 engagement data to understand what drives viewership across movies and TV shows. Using hours viewed and view counts as the primary engagement metrics, the analysis examines content type, global availability, release timing, and runtime across all titles that met Netflix's minimum reporting threshold.

Key findings:

- TV shows account for a disproportionately large share of total hours viewed relative to their title count, driven by long-running serialized content.
- Globally available titles significantly outperform non-global titles in both hours viewed and total views, regardless of content type.

- January through March releases generate the most cumulative viewership, suggesting Netflix’s biggest content drops are front-loaded in Q1.
- For movies, runtime has little correlation with engagement. For TV shows, longer cumulative runtimes (more episodes/seasons) are associated with higher total hours viewed.

2 Introduction

Netflix publishes a bi-annual engagement report disclosing viewership data for its titles. This transparency gives analysts a rare look into what audiences actually watch — not just what is trending. The dataset is sourced from TidyTuesday (July 2025), and includes view counts, hours viewed, release dates, runtimes, and global availability flags for both movies and TV shows covering January through June 2025.

This analysis follows the six phases of the data analysis process: **Ask, Prepare, Process, Analyze, Share, and Act.**

3 Ask: Business Task

Central question: What patterns in Netflix’s January–June 2025 engagement data can help explain content performance, and what do those patterns suggest about audience behavior and content strategy?

Supporting questions:

1. Which titles drove the most engagement by hours viewed?
 2. How do Movies and TV Shows compare in total hours viewed and view counts?
 3. Does global availability significantly impact engagement?
 4. How does release month relate to viewership volume?
 5. Is there a relationship between content runtime and hours viewed, and does it differ between movies and TV shows?
-

4 Prepare: Data Source

The data comes from Netflix’s official engagement report, compiled by the TidyTuesday project.

File	Description
<code>movies.csv</code>	Netflix movie titles with viewership metrics
<code>shows.csv</code>	Netflix TV show titles with viewership metrics

Columns (both files):

Column	Description
<code>source</code>	Source file name
<code>report</code>	Report period
<code>title</code>	Movie/Show title
<code>available_globally</code>	Whether the show/movie is available globally or not
<code>release_date</code>	Date the show/movie was released on Netflix
<code>hours_viewed</code>	Total hours viewed on the show/movie
<code>runtime</code>	Content runtime

Column	Description
views	Total number of hours viewed

Limitations: The report covers only Jan–Jun 2025 and excludes titles that did not reach a minimum viewership threshold. It does not include genre, language, or production budget data.

5 Process: Data Cleaning & Preparation

5.1 Load Libraries

```
library(tidyverse)
library(knitr)
library(scales)
```

5.2 Load Data

```
movies <- readr::read_csv(
  'https://raw.githubusercontent.com/rfordatascience/tidytuesday/main/data/2025/2025-07-29/movies.csv')

shows <- readr::read_csv(
  'https://raw.githubusercontent.com/rfordatascience/tidytuesday/main/data/2025/2025-07-29/shows.csv')
```

5.3 Initial Inspection

```
glimpse(movies)

## Rows: 36,121
## Columns: 8
## $ source      <chr> "1_What_We_Watched_A_Netflix_Engagement_Report_2025~
## $ report      <chr> "2025Jan-Jun", "2025Jan-Jun", "2025Jan-Jun", "2025J~
## $ title       <chr> "Back in Action", "STRAW", "The Life List", "Exterr~
## $ available_globally <chr> "Yes", "Yes", "Yes", "Yes", "Yes", "No", "Yes", "Ye~
## $ release_date <date> 2025-01-17, 2025-06-06, 2025-03-28, 2025-04-30, 20~
## $ hours_viewed <dbl> 313000000, 185200000, 198900000, 159000000, 1549000~
## $ runtime     <chr> "1H 54M 0S", "1H 48M 0S", "2H 5M 0S", "1H 49M 0S", ~
## $ views       <dbl> 164700000, 102900000, 95500000, 87500000, 86900000,~
```

```
glimpse(shows)

## Rows: 27,803
## Columns: 8
## $ source      <chr> "1_What_We_Watched_A_Netflix_Engagement_Report_2025~
## $ report      <chr> "2025Jan-Jun", "2025Jan-Jun", "2025Jan-Jun", "2025J~
## $ title       <chr> "Adolescence: Limited Series", "Squid Game: Season ~
## $ available_globally <chr> "Yes", "Yes", "Yes", "Yes", "Yes", "Yes", "Yes", "Y~
## $ release_date <date> 2025-03-13, 2024-12-26, 2025-06-27, 2025-02-20, 20~
## $ hours_viewed <dbl> 555100000, 840300000, 438600000, 315800000, 2186000~
## $ runtime     <chr> "3H 50M 0S", "7H 10M 0S", "6H 8M 0S", "5H 9M 0S", "~
## $ views       <dbl> 144800000, 117300000, 71500000, 61300000, 58000000,~
```

```
colSums(is.na(movies))
```

```
##           source           report           title available_globally
##           0             0             6             9
##  release_date    hours_viewed      runtime           views
##      29396             12             12             12
```

```
colSums(is.na(shows))
```

```
##           source           report           title available_globally
##           0             0             6             9
##  release_date    hours_viewed      runtime           views
##      14033             12             714             12
```

5.4 Parse Runtime to Minutes

The `runtime` column is stored as a string (e.g., "1H 54M"). We convert it to numeric minutes for analysis.

```
parse_runtime_minutes <- function(runtime_str) {
  hours   <- as.numeric(str_extract(runtime_str, "(?<=^|\\s)(\\d+)(?=H)"))
  minutes <- as.numeric(str_extract(runtime_str, "(\\d+)(?=M)"))
  hours   <- replace_na(hours, 0)
  minutes <- replace_na(minutes, 0)
  hours * 60 + minutes
}
```

5.5 Combine Datasets

```
movies_clean <- movies |>
mutate(
  content_type = "Movie",
  runtime_minutes = parse_runtime_minutes(runtime),
  release_date = as.Date(release_date),
  release_month = month(release_date, label = TRUE, abbr = TRUE),
  release_quarter = paste0("Q", quarter(release_date)),
  global = if_else(available_globally == "Yes", "Global", "Non-Global")
)

shows_clean <- shows |>
mutate(
  content_type = "TV Show",
  runtime_minutes = parse_runtime_minutes(runtime),
  release_date = as.Date(release_date),
  release_month = month(release_date, label = TRUE, abbr = TRUE),
  release_quarter = paste0("Q", quarter(release_date)),
  global = if_else(available_globally == "Yes", "Global", "Non-Global")
)

all_content <- bind_rows(movies_clean, shows_clean)
```

5.6 Quick Check

```
all_content |>
  count(content_type, global) |>
  kable(caption = "Title count by content type and availability")
```

Table 1: Title count by content type and availability

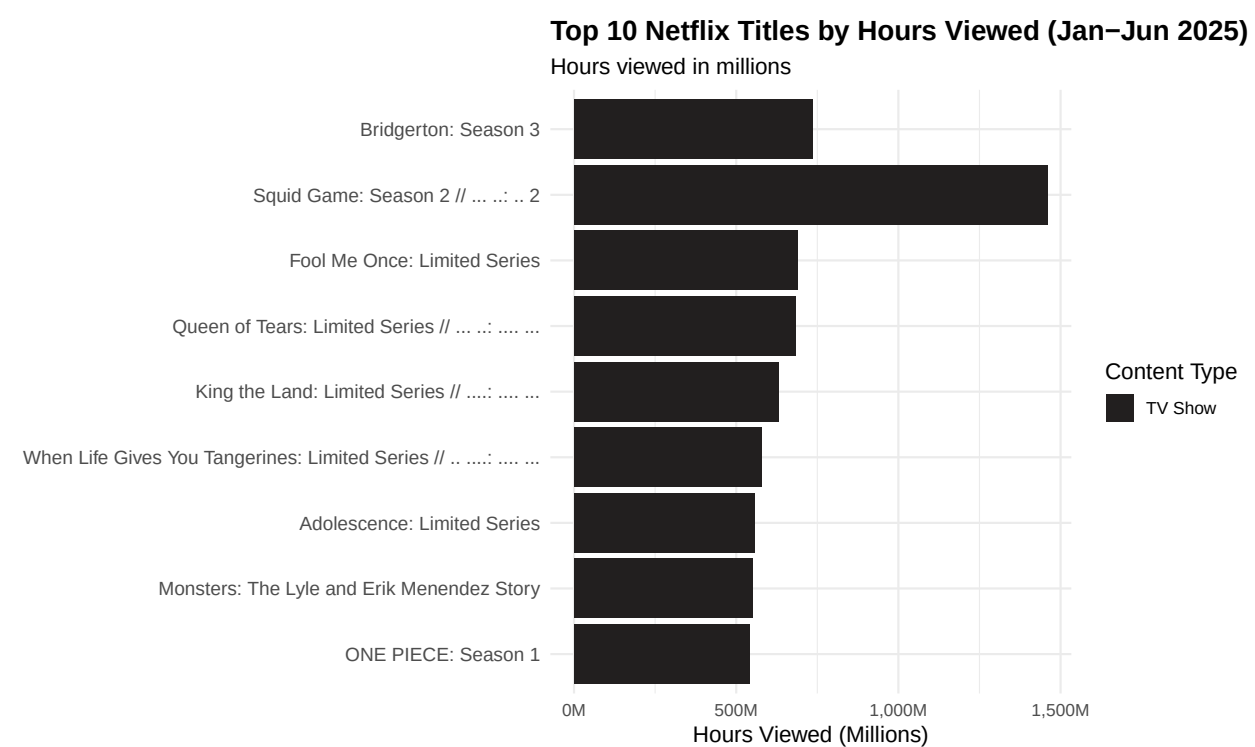
content_type	global	n
Movie	Global	6874
Movie	Non-Global	29238
Movie	NA	9
TV Show	Global	11089
TV Show	Non-Global	16705
TV Show	NA	9

6 Analyze: Exploration & Findings

6.1 Top 10 Titles by Hours Viewed

```
top_titles <- all_content |>
  slice_max(hours_viewed, n = 10) |>
  mutate(title = fct_reorder(title, hours_viewed))

ggplot(top_titles, aes(x = title, y = hours_viewed / 1e6, fill = content_type)) +
  geom_col() +
  coord_flip() +
  scale_fill_manual(values = c("Movie" = "#E50914", "TV Show" = "#221F1F")) +
  scale_y_continuous(labels = label_comma(suffix = "M")) +
  labs(
    title = "Top 10 Netflix Titles by Hours Viewed (Jan-Jun 2025)",
    subtitle = "Hours viewed in millions",
    x = NULL,
    y = "Hours Viewed (Millions)",
    fill = "Content Type"
  ) +
  theme_minimal(base_size = 13) +
  theme(
    plot.title = element_text(face = "bold"),
    axis.text.y = element_text(size = 11)
  )
```



6.2 Movies vs. TV Shows: Aggregate Engagement

```
type_summary <- all_content |>
  group_by(content_type) |>
  summarise(
    n_titles      = n(),
    total_hours   = sum(hours_viewed, na.rm = TRUE),
    total_views   = sum(views, na.rm = TRUE),
    median_hours  = median(hours_viewed, na.rm = TRUE),
    avg_hours_per_title = mean(hours_viewed, na.rm = TRUE)
  )

kable(type_summary |>
  mutate(across(where(is.numeric) & !n_titles, ~ comma(.))),
  caption = "Engagement summary: Movies vs. TV Shows")
```

Table 2: Engagement summary: Movies vs. TV Shows

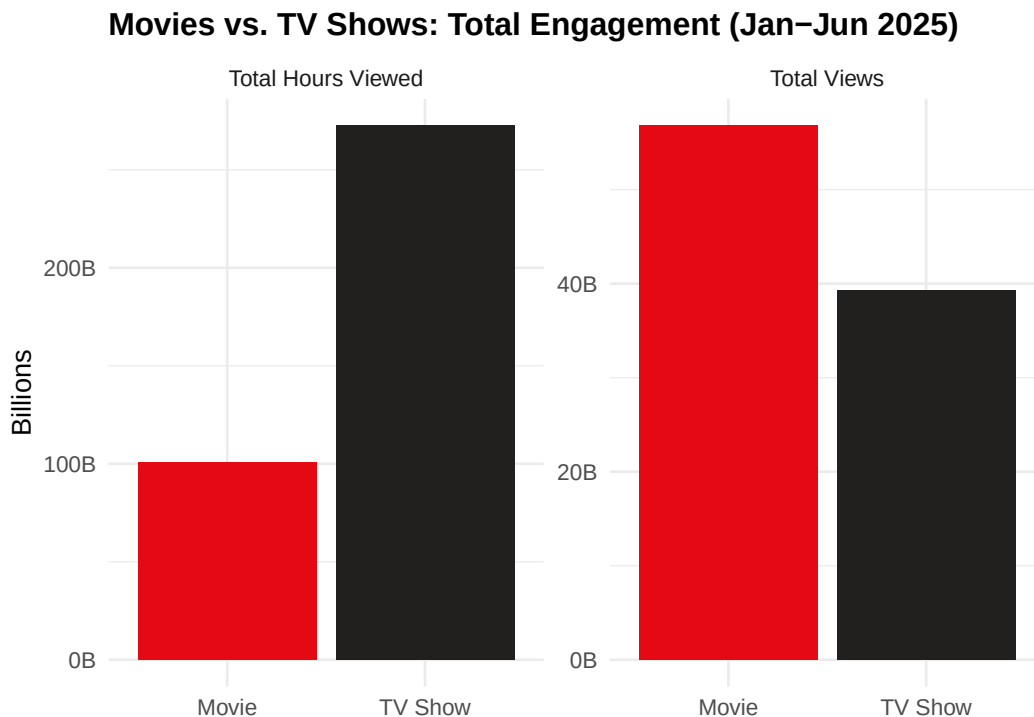
content_type	n_titles	total_hours	total_views	median_hours	avg_hours_per_title
Movie	36121	100,775,100,000	56,808,700,000	400,000	2,790,858
TV Show	27803	272,553,100,000	39,324,100,000	2,500,000	9,807,243

```
type_summary |>
  pivot_longer(cols = c(total_hours, total_views), names_to = "metric", values_to = "value") |>
  mutate(metric = recode(metric,
    total_hours = "Total Hours Viewed",
    total_views = "Total Views"
```

```

)) |>
ggplot(aes(x = content_type, y = value / 1e9, fill = content_type)) +
  geom_col(show.legend = FALSE) +
  facet_wrap(~metric, scales = "free_y") +
  scale_fill_manual(values = c("Movie" = "#E50914", "TV Show" = "#221F1F")) +
  scale_y_continuous(labels = label_comma(suffix = "B")) +
  labs(
    title = "Movies vs. TV Shows: Total Engagement (Jan-Jun 2025)",
    x = NULL,
    y = "Billions"
  ) +
  theme_minimal(base_size = 12) +
  theme(plot.title = element_text(face = "bold"))

```



6.3 Global vs. Non-Global Availability

```

global_summary <- all_content |>
  group_by(content_type, global) |>
  summarise(
    n_titles = n(),
    total_hours = sum(hours_viewed, na.rm = TRUE),
    median_views = median(views, na.rm = TRUE),
    .groups = "drop"
  )

kable(
  global_summary |>
    mutate(across(c(total_hours, median_views), ~ comma(.))),
  caption = "Engagement by content type and global availability"
)

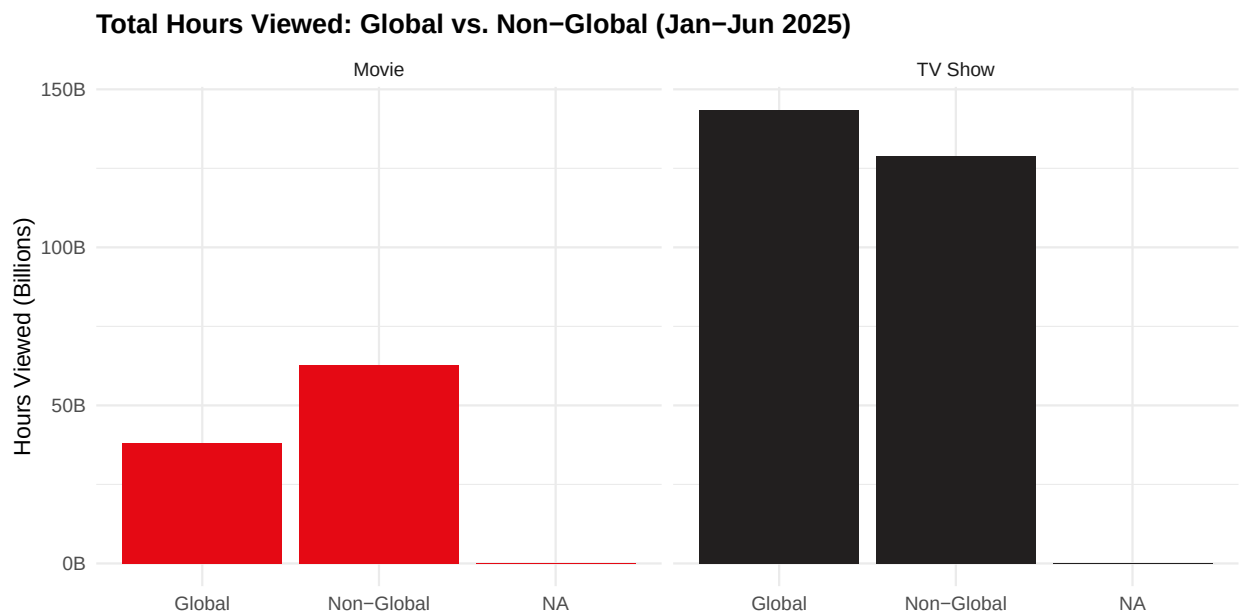
```

)

Table 3: Engagement by content type and global availability

content_type	global	n_titles	total_hours	median_views
Movie	Global	6874	38,022,700,000	700,000
Movie	Non-Global	29238	62,752,400,000	200,000
Movie	NA	9	0	NA
TV Show	Global	11089	143,594,900,000	800,000
TV Show	Non-Global	16705	128,958,200,000	300,000
TV Show	NA	9	0	NA

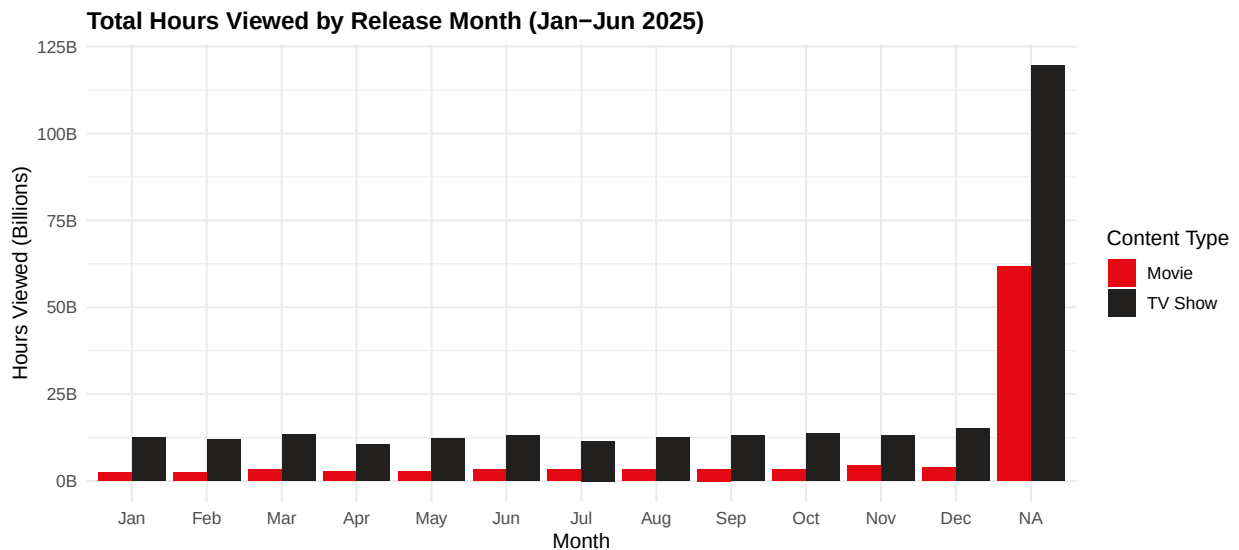
```
global_summary |>
  ggplot(aes(x = global, y = total_hours / 1e9, fill = content_type)) +
  geom_col(position = "dodge") +
  scale_fill_manual(values = c("Movie" = "#E50914", "TV Show" = "#221F1F")) +
  scale_y_continuous(labels = label_comma(suffix = "B")) +
  facet_wrap(~content_type) +
  labs(
    title = "Total Hours Viewed: Global vs. Non-Global (Jan-Jun 2025)",
    x = NULL,
    y = "Hours Viewed (Billions)",
    fill = "Content Type"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    plot.title = element_text(face = "bold"),
    legend.position = "none"
  )
```



6.4 Viewership by Release Month

```
monthly_summary <- all_content |>
  group_by(release_month, content_type) |>
  summarise(
    total_hours = sum(hours_viewed, na.rm = TRUE),
    n_titles    = n(),
    .groups = "drop"
  )

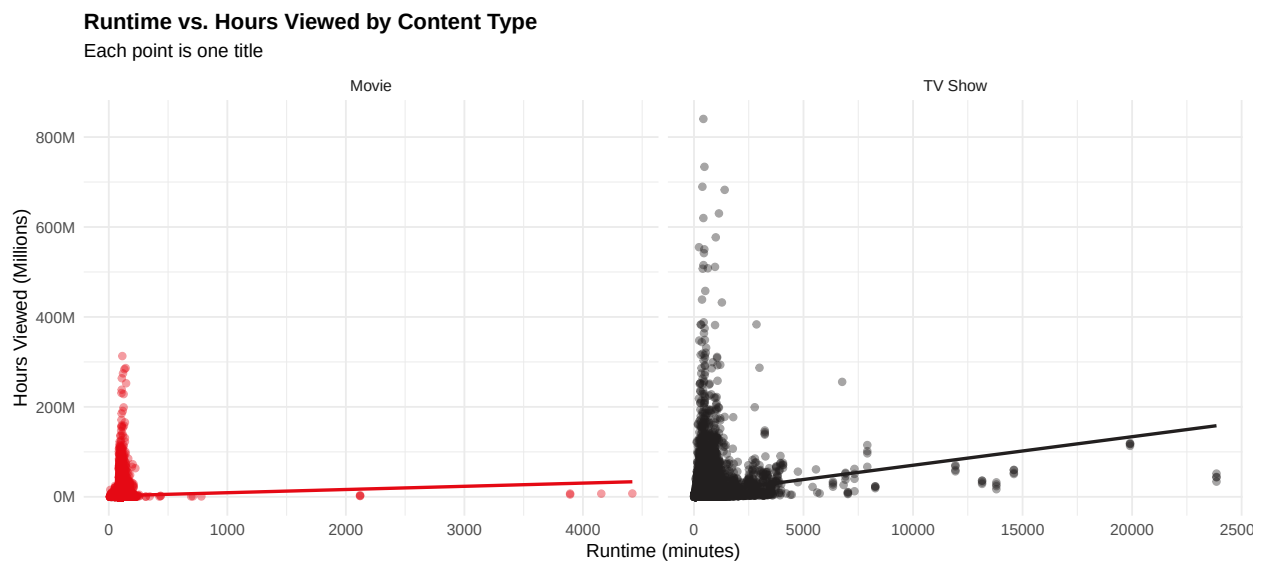
monthly_summary |>
  ggplot(aes(x = release_month, y = total_hours / 1e9, fill = content_type)) +
  geom_col(position = "dodge") +
  scale_fill_manual(values = c("Movie" = "#E50914", "TV Show" = "#221F1F")) +
  scale_y_continuous(labels = label_comma(suffix = "B")) +
  labs(
    title = "Total Hours Viewed by Release Month (Jan-Jun 2025)",
    x      = "Month",
    y      = "Hours Viewed (Billions)",
    fill   = "Content Type"
  ) +
  theme_minimal(base_size = 12) +
  theme(plot.title = element_text(face = "bold"))
```



6.5 Runtime vs. Engagement

```
all_content |>
  filter(runtime_minutes > 0, hours_viewed > 0) |>
  ggplot(aes(x = runtime_minutes, y = hours_viewed / 1e6,
             color = content_type)) +
  geom_point(alpha = 0.4, size = 1.8) +
  geom_smooth(method = "lm", se = FALSE, linewidth = 1) +
  facet_wrap(~content_type, scales = "free_x") +
  scale_color_manual(values = c("Movie" = "#E50914", "TV Show" = "#221F1F")) +
  scale_y_continuous(labels = label_comma(suffix = "M")) +
```

```
labs(
  title = "Runtime vs. Hours Viewed by Content Type",
  subtitle = "Each point is one title",
  x = "Runtime (minutes)",
  y = "Hours Viewed (Millions)",
  color = "Content Type"
) +
theme_minimal(base_size = 12) +
theme(
  plot.title = element_text(face = "bold"),
  legend.position = "none"
)
```



7 Share: Key Findings

Q1 — Top titles by hours viewed: The highest-performing titles are dominated by TV shows, with a small number of breakout movies. A handful of titles account for a disproportionate share of total platform hours, illustrating a long-tail distribution in engagement.

Q2 — Movies vs. TV Shows: TV shows generate more total hours viewed than movies despite movies making up a larger share of total titles. On a per-title basis, however, top movies can compete with or exceed TV shows in view counts, suggesting movies drive discrete spikes while TV sustains ongoing consumption.

Q3 — Global availability: Globally available titles substantially outperform non-global titles in both hours viewed and total views across both content types. The gap is especially pronounced for TV shows, where non-global titles have a much lower median.

Q4 — Release timing: Q1 (January–March) releases accumulate the most total hours viewed. January in particular stands out, consistent with Netflix releasing tent-pole content at the start of the year to capitalize on post-holiday viewership.

Q5 — Runtime vs. engagement: For movies, runtime shows little to no linear relationship with hours viewed — a 90-minute film can perform as well as a 2.5-hour one. For TV shows, longer cumulative runtimes correlate positively with total hours viewed, reflecting how serialized content compounds engagement across episodes and seasons.

8 Act: Recommendations

1. **Prioritize global rollouts** for any major release. The data consistently shows globally available titles outperform non-global ones — restricting availability is a measurable engagement cost.
 2. **Front-load the release calendar.** Q1, especially January, is the highest viewership window. Flagship content should target this window rather than being spread evenly across the year.
 3. **Invest in serialized TV.** Multi-season shows compound hours viewed in a way individual films cannot. Even mid-tier TV shows accumulate more total hours than all but the top-performing movies.
 4. **Don't optimize movie runtime for engagement.** Since runtime and hours viewed are uncorrelated for films, runtime decisions should be driven by storytelling, not a belief that longer movies perform better.
-

9 Tableau Export

The following code exports cleaned CSV files that can be imported directly into Tableau for interactive dashboard building.

```
dir.create("../data/tableau", showWarnings = FALSE, recursive = TRUE)

# Full cleaned dataset
write_csv(all_content, "../data/tableau/all_content.csv")

# Monthly summary
monthly_export <- all_content |>
  group_by(release_month, release_quarter, content_type) |>
  summarise(
    n_titles = n(),
    total_hours = sum(hours_viewed, na.rm = TRUE),
    total_views = sum(views, na.rm = TRUE),
    avg_hours = mean(hours_viewed, na.rm = TRUE),
    .groups = "drop"
  )
write_csv(monthly_export, "../data/tableau/monthly_summary.csv")

# Global vs non-global summary
write_csv(global_summary, "../data/tableau/global_summary.csv")

# Top 50 titles
top_50 <- all_content |>
  slice_max(hours_viewed, n = 50) |>
  select(title, content_type, global, release_date, release_month,
         release_quarter, runtime_minutes, hours_viewed, views)
write_csv(top_50, "../data/tableau/top_50_titles.csv")

# Content type totals
write_csv(type_summary, "../data/tableau/type_summary.csv")

cat("Exported 5 CSV files to data/tableau/\n")
```

```
## Exported 5 CSV files to data/tableau/
```